Extra Practice 1: Angles in Standard Position and Reference Angles

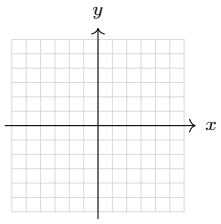
A. Skills

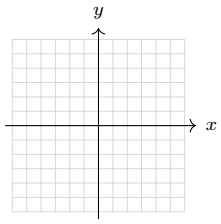
1. State the quadrant in which each angle terminates, or the axis it lies on.
(a) 199° (b) 89° (c) 301° (d) 91° (e) 270° (f) 179°

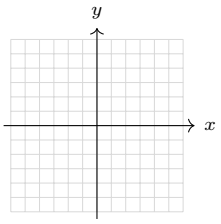
2. Determine the reference angle of each angle.
(a) 152° (b) 212° (c) 299° (d) 76° (e) 180° (f) 263°

3. Determine the rotation angle θ, $0° \leq \theta \leq 360°$, from its reference angle and quadrant.
(a) $\theta_R = 27°$, quadrant 4 (b) $\theta_R = 54°$, quadrant 2 (c) $\theta_R = 89°$, quadrant 3 (d) $\theta_R = 3°$, quadrant 3

4. Each point is on the terminal arm of an angle θ in standard position. Plot the point, draw θ, and state its quadrant.
(a) $(-1, 5)$ (b) $(6, -6)$ (c) $(-5, -3)$







B. Applying

5. Determine the three other angles from 0° to 360° with the same reference angle as the given angle.
(a) 98° (b) 264° (c) 349°

6. State the angle from 0° to 360° that has the same terminal arm as each rotation.
(a) −310° (b) 500° (c) −90°

7. The point $P(8, 3)$ is on the terminal arm of an angle of 21° (to the nearest degree). State the image of P and the rotation angle through it after each reflection.
(a) in the y-axis (b) in both axes (c) in the x-axis

8. A student says the reference angle of $341°$ is $71°$. Use a sketch to explain the mistake and give the correct reference angle.

9. Which angle does *not* have a reference angle of $25°$?

A. $155°$ **B.** $205°$ **C.** $245°$ **D.** $335°$

10. Numerical Response An angle θ terminates in quadrant 3 and has a reference angle of $38°$. The value of θ is _____°.

C. Stretch

11. An angle θ terminates in quadrant 2. Write an expression in terms of θ for the angle in quadrant 4 with the same reference angle. Check your expression with $\theta = 130°$.

12. Angle A terminates in quadrant 2 and angle B terminates in quadrant 4. They have the same reference angle, and $A + B = 400°$. Determine A and B.

13. Show that the four angles from $0°$ to $360°$ with any reference angle x (where $0° < x < 90°$) always add to $720°$.

Extra Practice 2: Trigonometric Ratios from a Point on the Terminal Arm

A. Skills

1. Determine the exact distance r from the origin to each point.
(a) $(-10, 24)$ (b) $(3, -3)$ (c) $(-1, -7)$ (d) $(6, 0)$

2. Each point is on the terminal arm of θ. Determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ (rational denominators).
(a) $(35, 12)$ (b) $(-16, -30)$ (c) $(4, -2)$

3. State each value. Write "undefined" where it applies.
(a) $\sin 270°$ (b) $\cos 180°$ (c) $\tan 90°$ (d) $\tan 180°$ (e) $\cos 90°$

4. Each point is on the terminal arm of θ, $0° \leq \theta < 360°$. Determine θ to the nearest tenth of a degree.
(a) $(2, 5)$ (b) $(-8, 3)$ (c) $(-1, -4)$ (d) $(10, -7)$

B. Applying

5. Write each ratio as the same ratio of its reference angle, with the correct sign.
(a) $\cos 200°$ (b) $\sin 305°$ (c) $\tan 135°$ (d) $\sin 170°$

6. A ship is at $(30, -16)$ on a map grid with the port at the origin, east along the positive x-axis and north along the positive y-axis (units: km).
(a) How far is the ship from the port?

(b) Determine the exact values of $\sin\theta$ and $\cos\theta$, where θ is the rotation angle from east to the ship.

(c) Determine θ to the nearest tenth of a degree.

7. Without calculating any values, state the sign of $\sin\theta$, $\cos\theta$ and $\tan\theta$ when $(-7, 3)$ is on the terminal arm.

8. For the point $(-3, -4)$, a student wrote $r = -5$ and $\sin\theta = \dfrac{-4}{-5} = \dfrac{4}{5}$. Explain the error and correct it.

9. The point $(-2, -6)$ is on the terminal arm of θ. The exact value of $\sin\theta$ is

A. $-\dfrac{3\sqrt{10}}{10}$ **B.** $\dfrac{3\sqrt{10}}{10}$ **C.** $-\dfrac{\sqrt{10}}{10}$ **D.** 3

10. Numerical Response The point $(-7, 7)$ is on the terminal arm of θ, $0° \leq \theta < 360°$. The value of θ is _____°.

C. Stretch

11. The point $(a, 2a)$, with $a < 0$, is on the terminal arm of θ. Determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$, and θ to the nearest tenth of a degree.

12. The point $(k, 4)$ is on the terminal arm of θ, where $r = 5$ and θ terminates in quadrant 2. Determine k, the exact values of the three ratios, and θ to the nearest tenth of a degree.

13. Use $x^2 + y^2 = r^2$ to show that $(\sin\theta)^2 + (\cos\theta)^2 = 1$ for every angle θ. Check it with the point $(-20, 21)$.

Extra Practice 3: The CAST Rule: Signs and Solving for Angles

A. Skills

1. Name the quadrant in which θ terminates.
(a) $\tan\theta < 0$ and $\cos\theta > 0$ _____ (b) $\sin\theta > 0$ and $\cos\theta > 0$ _____ (c) $\tan\theta > 0$ and $\cos\theta < 0$ _____

2. Without a calculator, state whether each ratio is positive or negative.
(a) $\cos 104°$ _____ (b) $\tan 263°$ _____ (c) $\sin 333°$ _____ (d) $\cos 352°$ _____

3. Write each ratio as the same ratio of its reference angle, with the correct sign.
(a) $\sin 172°$ _____ (b) $\cos 316°$ _____ (c) $\tan 125°$ _____ (d) $\sin 247°$ _____

4. Solve for θ, $0° \le \theta < 360°$, to the nearest tenth of a degree.
(a) $\sin\theta = 0.62$ (b) $\cos\theta = -0.12$ (c) $\tan\theta = 0.25$

(d) $\sin\theta = -0.95$ (e) $\cos\theta = 0.5$ (f) $\tan\theta = -6$

5. Solve for θ, $0° \le \theta < 360°$. (a) $\sin\theta = 1$ (b) $\cos\theta = 2.1$ (c) $\tan\theta = 0$ (d) $\sin\theta = 0$

B. Applying

6. Given $\sin\theta = -\frac{7}{25}$ and $\cos\theta < 0$, determine the exact values of $\cos\theta$ and $\tan\theta$.

7. Given $\tan\theta = \frac{5}{2}$ and $\cos\theta < 0$, determine the exact values of $\sin\theta$ and $\cos\theta$ with rational denominators.

8. Solve $3\tan\theta - 5 = 2\tan\theta - 1$ for $0° \le \theta < 360°$, to the nearest tenth of a degree.

9. The end of a lawn sprinkler arm is $x = 2.5\cos\theta$ metres east of its pivot after the arm rotates θ from due east (negative means west). For $0° \le \theta < 360°$, at what angles is the end 2 m west of the pivot? Answer to the nearest tenth of a

degree.

10. Which ratio has the same value as $\tan 128°$?

A. $\tan 52°$ **B.** $-\tan 52°$ **C.** $\tan 232°$ **D.** $-\tan 308°$

11. Numerical Response For $0° \leq \theta \leq 180°$, the solution of $\cos\theta = -0.85$, to the nearest degree, is _____°.

C. Stretch

12. Solve $(2\sin\theta - 1)(5\cos\theta - 1) = 0$ for $0° \leq \theta < 360°$, to the nearest tenth of a degree. (A product is zero when one factor is zero.)

13. Angle θ is in Quadrant IV with $\cos\theta = k$, where $0 < k < 1$. Write $\sin\theta$ and $\tan\theta$ in terms of k.

14. For which angles θ, $0° \leq \theta < 360°$, is $\sin\theta = \cos\theta$? (Hint: on the terminal arm, $\frac{y}{r} = \frac{x}{r}$ means $y = x$.)

Extra Practice 4: Exact Values from Special Triangles

A. Skills

1. Determine the exact value.

(a) $\sin 330°$ (b) $\cos 315°$ (c) $\tan 120°$ (d) $\cos 180°$ (e) $\sin 135°$

(f) $\tan 240°$ (g) $\cos 210°$ (h) $\sin 90°$ (i) $\tan 300°$ (j) $\cos 270°$

2. Solve for θ, $0° \leq \theta < 360°$, using exact values.

(a) $\sin\theta = \frac{1}{2}$ (b) $\cos\theta = \frac{\sqrt{3}}{2}$ (c) $\tan\theta = 1$

(d) $\cos\theta = 0$ (e) $\sin\theta = -1$ (f) $\tan\theta = \frac{\sqrt{3}}{3}$

3. Each point is on the terminal arm of θ. Determine θ exactly, $0° \leq \theta < 360°$.

(a) $(-7, 7)$ (b) $(\sqrt{3}, -3)$ (c) $(-4\sqrt{3}, 4)$

B. Applying

4. Determine the exact value of each expression.

(a) $\cos^2 30° - \sin^2 30°$ (b) $\tan 225° + \cos 180°$ (c) $6 \sin 300° \tan 150°$ (d) $\frac{\sin 45°}{\cos 315°}$

5. Solve for θ, $0° \leq \theta < 360°$, using exact values. (a) $4\sin^2\theta - 3 = 0$ (b) $2\cos^2\theta = 1$

6. Solve in the given interval, using exact values. (a) $2\cos\theta = -1$, $0° \leq \theta \leq 180°$ (b) $\tan\theta + 1 = 0$, $180° \leq \theta < 360°$

7. Given $\sin A = -\dfrac{\sqrt{3}}{2}$ and $\tan A > 0$, determine A $(0° \leq A < 360°)$ and the exact values of $\cos A$ and $\tan A$.

8. The solutions of $\sqrt{2}\sin\theta - 1 = 0$, $0° \leq \theta < 360°$, are

A. $45°, 135°$ **B.** $45°, 315°$ **C.** $135°, 225°$ **D.** $45°, 225°$

9. Numerical Response The exact value of $12\cos^2 30° - \tan 45°$ is _____.

C. Stretch

10. A point moves counterclockwise around a circle centred at the origin, from $A(-3\sqrt{3}, 3)$ to $B(3, -3\sqrt{3})$. Through what angle does it rotate? What is the radius of the circle?

11. Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for $0° \leq \theta < 360°$. (Hint: let $s = \sin\theta$ and factor $2s^2 - s - 1$.)

12. Show that $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 150°$ and $\theta = 315°$. Then use x, y and r to explain why it is true for every angle.

Extra Practice 5: The Sine Law

Round sides to the nearest tenth and angles to the nearest degree unless a question says otherwise.

A. Skills

1. Sketch each triangle and determine the indicated side.
(a) $\angle A = 39°$, $\angle B = 82°$, $a = 10.0$ cm; find b.
(b) $\angle P = 118°$, $\angle Q = 27°$, $p = 30.0$ m; find q and r.
(c) $\angle J = 47°$, $\angle L = 68°$, $j = 52$ km; find k.

2. Determine the indicated angle. Explain in a few words why only one triangle is possible.
(a) $\angle A = 76°$, $a = 18.5$, $b = 14.2$; find $\angle B$.
(b) $\angle R = 131°$, $r = 25.4$, $q = 16.9$; find $\angle Q$.
(c) Acute $\triangle XYZ$: $\angle X = 62°$, $x = 11.0$, $y = 12.0$; find $\angle Y$ and $\angle Z$.

3. For each set of information about $\triangle ABC$, decide whether the sine law can be used as the first step. If it can, write the equation you would solve (do not solve it).
(a) $\angle B = 50°$, $b = 9$, $\angle C = 72°$; find c.
(b) $a = 7$, $b = 10$, $\angle C = 45°$; find c.
(c) $\angle A = 35°$, $\angle B = 95°$, $c = 12$; find a.
(d) $a = 6$, $b = 8$, $c = 11$; find $\angle C$.

4. Without a calculator, name the acute angle with the same sine as each angle. (a) $\sin 145°$ (b) $\sin 98°$ (c) $\sin 120°$

B. Applying

5. In $\triangle ABC$, $\angle A = 57°$, $a = 30.6$ cm and $b = 22.4$ cm. Solve the triangle.

6. Two observers stand 2.5 km apart on level ground with a hot-air balloon between them. The angles of elevation of the balloon are 35° from one observer and 52° from the other. Sketch a diagram and determine the height of the balloon to the nearest hundredth of a kilometre.

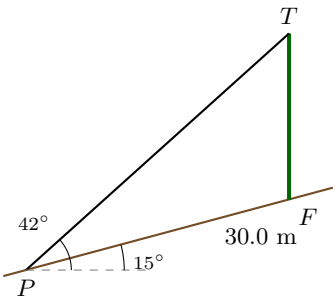
7. A roof truss spans 12.0 m. One side of the roof rises at 32° and the other at 45°. Determine the length of each sloped side.

8. Ranger station S is 24.0 km due east of ranger station R. A campfire F is on a bearing of N63°E from R and N18°E from S. Sketch a diagram and determine the distance from each station to the fire.

9. In $\triangle ABC$, $\angle B = 48°$, $\angle C = 21°$ and $b = 9.3$ cm. To the nearest tenth, a is

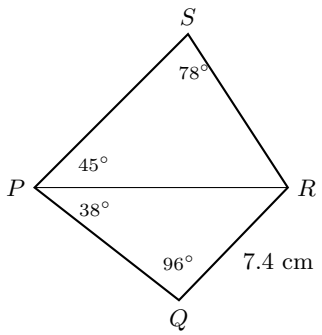
A. 4.5 cm **B.** 7.4 cm **C.** 11.7 cm **D.** 19.3 cm

10. Numerical Response A vertical tree FT grows on a hillside that slopes at 15°. From point P, 30.0 m downhill from the foot F, the angle of elevation of the top T (from the horizontal) is 42°. The height of the tree, to the nearest tenth of a metre, is _____ m.



C. Stretch

11. Quadrilateral $PQRS$ is split by the diagonal PR. In $\triangle PQR$, $\angle Q = 96°$, $\angle QPR = 38°$ and $QR = 7.4$ cm. In $\triangle PRS$, $\angle RPS = 45°$ and $\angle S = 78°$. Determine RS.



12. Use the sine law to show that $\dfrac{a}{b} = \dfrac{\sin A}{\sin B}$ in any $\triangle ABC$. Then explain why, if $a > b$, then $\angle A > \angle B$ (the longer side is opposite the larger angle) when both angles are acute.

Extra Practice 6: The Cosine Law

Round sides to the nearest tenth and angles to the nearest degree unless a question says otherwise.

A. Skills

1. Sketch each triangle and determine the third side.
(a) $\angle C = 47°$, $a = 14$ cm, $b = 9$ cm; find c. (b) $\angle B = 138°$, $a = 3.6$ m, $c = 5.1$ m; find b. (c) $\angle A = 29°$, $b = 120$ km, $c = 85$ km; find a.

2. Determine the indicated angle.
(a) $a = 11$, $b = 13$, $c = 15$; find $\angle A$. (b) $a = 12$, $b = 16$, $c = 25$; find $\angle C$. (c) $a = 4.4$, $b = 7.2$, $c = 9.0$; find $\angle A$.

3. Without a calculator, determine a exactly in $\triangle ABC$ if $\angle A = 30°$, $b = 2\sqrt{3}$ and $c = 4$.

4. Write the equation you would use, but do not solve it. (a) In $\triangle DEF$, find e given d, f and $\angle E$. (b) In $\triangle XYZ$, find $\angle Y$ given x, y and z.

B. Applying

5. Solve the triangle with sides $a = 9.4$ cm, $b = 12.7$ cm and $c = 15.1$ cm.

6. A tunnel will be built straight through a hill from A to B. From a point C, a surveyor measures $CA = 386$ m, $CB = 512$ m and $\angle ACB = 71°$. Determine the length of the tunnel to the nearest metre.

7. An 8.5 m vertical pole stands on a hillside that slopes at 14°. A guy wire runs from the top of the pole to a stake 12.0 m downhill from the base of the pole. Sketch a diagram and determine the length of the wire.

8. A clock has an hour hand 6 cm long and a minute hand 9 cm long. Determine the distance between the tips of the hands at 4:00, exactly and to the nearest tenth.

9. The largest angle of a triangle with sides 8, 13 and 15, to the nearest tenth of a degree, is

A. 32.2° **B.** 60.0° **C.** 87.8° **D.** 92.2°

10. Numerical Response Two straight paths lead from a fountain to two benches, 9.0 m and 11.5 m away. The angle between the paths is 74°. The distance between the benches, to the nearest tenth of a metre, is _____ m.

C. Stretch

11. A box measures 8 cm by 6 cm by 5 cm. From one corner, a diagonal is drawn across each of the three faces that meet there; their far ends form a triangle. Determine the side lengths of that triangle (exactly) and its largest angle.

12. Show, without a calculator, that a triangle with sides 7, 8 and 13 has a 120° angle.

Extra Practice 7: The Ambiguous Case

A. Skills

1. In $\triangle ABC$, $\angle A = 50°$ and $b = 16$ cm. Determine the height $h = b\sin A$ to the nearest tenth, then state how many triangles are possible for each length of a.
(a) $a = 11$ cm (b) $a = 14$ cm (c) $a = 19$ cm

2. Without using the sine law, decide whether there are 0, 1 or 2 triangles.
(a) $\triangle RST$: $\angle S = 57°$, $s = 7.5$, $r = 6.1$ (b) $\triangle UVW$: $\angle U = 24°$, $u = 4.8$, $v = 10.5$ (c) $\triangle GHK$: $\angle G = 100°$, $g = 7.3$, $h = 8.1$

3. Determine all possible measures of $\angle B$, to the nearest degree, in $\triangle ABC$.
(a) $\angle A = 44°$, $a = 13$, $b = 16$ (b) $\angle A = 75°$, $a = 9.2$, $b = 8.4$ (c) $\angle A = 52°$, $a = 5$, $b = 7.5$

B. Applying

4. Solve all possible triangles ABC with $\angle A = 36°$, $a = 15$ m and $b = 21$ m. Round angles to the nearest degree and sides to the nearest tenth.

5. In $\triangle DEF$, $\angle D = 131°$, $d = 25$ cm and $e = 16$ cm. Solve the triangle (angles to the nearest degree, sides to the nearest tenth).

6. A radar station R is 10 km from a straight highway entrance E. The highway makes an angle of 29° with line ER. The radar detects vehicles up to 6 km away. Between what distances from E (to the nearest tenth of a km) along the highway is a car within radar range?

7. A robot arm has a fixed upper link PQ of length 2.2 m that makes an angle of $48°$ with a straight conveyor belt at P. A 1.8 m lower link hinged at Q swings down until its tip touches the belt. Find both possible distances from P to the tip, to the nearest tenth of a metre.

8. In $\triangle ABC$, $\angle A = 52°$ and $b = 18$ cm. For what lengths of a (nearest tenth) is there exactly one triangle?

9. In $\triangle ABC$, $\angle A = 54°$, $a = 8.5$ cm and $b = 10$ cm. How many triangles are possible?

A. none

B. exactly one

C. exactly two

D. exactly one, and it is a right triangle

10. Numerical Response In $\triangle ABC$, $\angle A = 27°$, $a = 5$ cm and $b = 9$ cm. To the nearest tenth of a centimetre, the shorter possible length of side c is _____ cm.

C. Stretch

11. In $\triangle ABC$, $\angle A = 30°$ and $b = 10$ cm. Find the length of a, to the nearest hundredth, for which the two possible values of $\angle B$ differ by $60°$.

12. Show that when $a = h = b \sin A$, the sine law gives exactly one angle B, and say what kind of triangle results.

13. In $\triangle PQR$, $\angle P = 40°$ and $PR = 12$ cm. Side QR can swing so that Q lands in two places on the other arm, 6 cm apart. Find the length of QR to the nearest tenth.

Extra Practice 8: Choosing the Sine Law or the Cosine Law

A. Skills

1. Name the most appropriate method to find the unknown, and justify it. Do not solve.
(a) $\triangle ABC$: $\angle A = 72°$, $b = 9$, $c = 14$; find a (b) $\triangle DEF$: $\angle D = 38°$, $\angle E = 77°$, $f = 20$; find d

(c) $\triangle GHK$: $\angle K = 90°$, $g = 5$, $h = 12$; find $\angle G$ (d) $\triangle LMN$: $l = 6.5$, $m = 4.2$, $n = 8.8$; find $\angle N$

2. Write each bearing in the other form. (a) 075° (b) 200° (c) S 50° E (d) N 35° W
(e) Q is on a bearing of 300° from P. Find the bearing of P from Q.

3. Find each area, to the nearest tenth.
(a) Sides 9 cm and 13 cm with a contained angle of 52°. (b) Sides 6 m, 8 m and 11 m. (c) An equilateral triangle with sides 10 cm.

4. A clock's minute hand is 10 cm long and its hour hand is 7 cm long. Find the angle between the hands at 11:00 and at 3:30, and the distance between the tips at 3:30 (nearest tenth).

B. Applying

5. A hiker walks 3.2 km on a bearing of 140° from camp P to Q, then 4.5 km on a bearing of 250° to R. Find the distance (nearest tenth of a km) and the bearing (nearest degree) from R back to camp.

6. A boat at sea sees the top of a cliff at an angle of elevation of 18°. After the boat moves 150 m straight toward the cliff, the angle of elevation is 31°. Find the height of the cliff, to the nearest metre.

7. Points A, B and C lie on a circle with centre O and radius 12 cm, and $\angle ACB = 35°$ (C on the major arc). Find $\angle AOB$ and the length of chord AB, to the nearest tenth.

8. A field $WXYZ$ is split by the diagonal WY. $WX = 85$ m, $XY = 60$ m, $\angle WXY = 120°$, $\angle YWZ = 30°$ and $\angle WYZ = 65°$. Find WY, WZ, YZ and the area of the field (sides to the nearest metre, area to the nearest 10 m^2).

9. A direction of S 20° W is the same as a bearing of

A. 020° **B.** 160° **C.** 200° **D.** 250°

10. Numerical Response Two boats leave a dock at the same time, one on a bearing of 310° at 22 km/h and the other on a bearing of 045° at 30 km/h. To the nearest kilometre, their distance apart after 40 minutes is _____ km.

C. Stretch

11. A regular pentagon is drawn inside a circle of radius 10 cm, with all five vertices on the circle. Find the side length and the area of the pentagon, to the nearest tenth.

12. A ship at P sails at 15 km/h on a bearing of 080°. A lighthouse L is 12 km from P on a bearing of 040°. A sailor can see the lighthouse's light from up to 9 km away. After how many minutes does the light come into view, and after how many minutes is it lost? (Nearest minute.) What is the closest the ship comes to the lighthouse?

Extra Practice 9: Unit 3 Review — Practice Test

Questions 1–6 ask for exact answers. Use Degree mode. Sides to the nearest tenth and angles to the nearest degree unless a question says otherwise.

A. Multiple Choice

1. An angle in standard position terminates in quadrant III and has a reference angle of 28°. The angle is

A. 152° **B.** 208° **C.** 242° **D.** 332°

2. Which statement is true?

A. $\cos 160° = \cos 20°$ **B.** $\sin 250° = -\sin 70°$ **C.** $\tan 130° = \tan 50°$ **D.** $\sin 310° = \sin 50°$

3. The point $(12, -5)$ is on the terminal arm of angle θ. The exact value of $\cos\theta$ is

A. $\frac{12}{13}$ **B.** $-\frac{12}{13}$ **C.** $\frac{5}{13}$ **D.** $-\frac{5}{13}$

4. The exact value of $\tan 300°$ is

A. $-\sqrt{3}$ **B.** $\sqrt{3}$ **C.** $-\frac{\sqrt{3}}{3}$ **D.** $-\frac{1}{2}$

5. If $\sin\theta = \frac{5}{13}$ and $90° < \theta < 180°$, then $\tan\theta$ is

A. $-\frac{5}{12}$ **B.** $\frac{5}{12}$ **C.** $-\frac{12}{5}$ **D.** $-\frac{12}{13}$

6. The largest solution of $2\cos\theta - \sqrt{2} = 0$, $0° \le \theta < 360°$, is

A. 45° **B.** 135° **C.** 225° **D.** 315°

7. The solutions of $\tan\theta = 0.9$, $0° \le \theta < 360°$, to the nearest tenth of a degree, are

A. $42.0°, 138.0°$ **B.** $42.0°, 222.0°$ **C.** $42.0°, 318.0°$ **D.** $138.0°, 318.0°$

8. In $\triangle PQR$, $\angle P = 27°$ and $q = 20$ cm. For which value of p is there exactly one triangle?

A. $p = 7$ cm **B.** $p = 12$ cm **C.** $p = 15$ cm **D.** $p = 25$ cm

9. A circle with centre O has a radius of 7.5 cm. Chord AB is 11 cm long. The measure of $\angle AOB$, to the nearest degree, is

A. 47° **B.** 86° **C.** 94° **D.** 133°

10. In $\triangle ABC$, $\angle A = 63°$, $a = 14.0$ cm and $c = 9.5$ cm. The measure of $\angle C$ is

A. 37° **B.** 143° **C.** 37° or 143° **D.** 53°

B. Numerical Response

11. Numerical Response The terminal arm of angle θ, $0° \leq \theta < 360°$, passes through $(-4, -9)$. To the nearest degree, θ is _____°.

12. Numerical Response A ship sails 42 km on a bearing of 035°, then turns and sails 30 km on a bearing of 120°. To the nearest kilometre, the ship is now _____ km from its starting point.

13. Numerical Response Points A and B are 300 m apart on level ground. A hot-air balloon is directly above a point between them. The angles of elevation of the balloon from A and B are 42° and 57°. To the nearest metre, the balloon is _____ m above the ground.

14. Numerical Response The larger solution of $4\cos\theta + 1 = 0$, $0° \leq \theta < 360°$, to the nearest degree, is _____°.

C. Written Response

15. The point $P(-2, 5)$ is on the terminal arm of angle θ.

(a) Sketch θ in standard position. Determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

(b) Determine θ to the nearest tenth of a degree.

(c) P is reflected in the origin to $Q(2, -5)$. Determine the angle in standard position whose terminal arm passes through Q, and explain why sin of this angle is the opposite of $\sin\theta$.

16. Exact answers.

(a) Given $\cos\theta = -\frac{2}{3}$ and $\tan\theta < 0$, determine the exact values of $\sin\theta$ and $\tan\theta$.

(b) Solve exactly for $0° \leq \theta < 360°$: (i) $2\cos\theta - 1 = 0$ (ii) $\tan^2\theta = 3$

(c) Evaluate exactly: $\tan 120° \cos 150° + \sin^2 45°$

17. In $\triangle ABC$, $\angle A = 42°$ and $b = 15$ cm. The length of side a can vary.

(a) For what values of a are there two triangles? (Give the boundary to the nearest hundredth.)

(b) Let $a = 12$ cm. Solve both triangles.

(c) Determine the difference between the areas of the two triangles in (b), to the nearest tenth of a square centimetre.

18. Points A and B are 85 m apart on one bank of a straight river. A tree C stands on the opposite bank. $\angle CAB = 58°$ and $\angle CBA = 74°$.

(a) Sketch the situation. Determine AC, and justify your choice of law.

(b) Determine the width of the river.

(c) Determine the area of $\triangle ABC$ to the nearest square metre.